

CCDS25138 - When Do Native Speech-Language Models Beat Cascades?

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Introduction

Voice agents are increasingly replacing traditional IVRs in customer support and telecom.



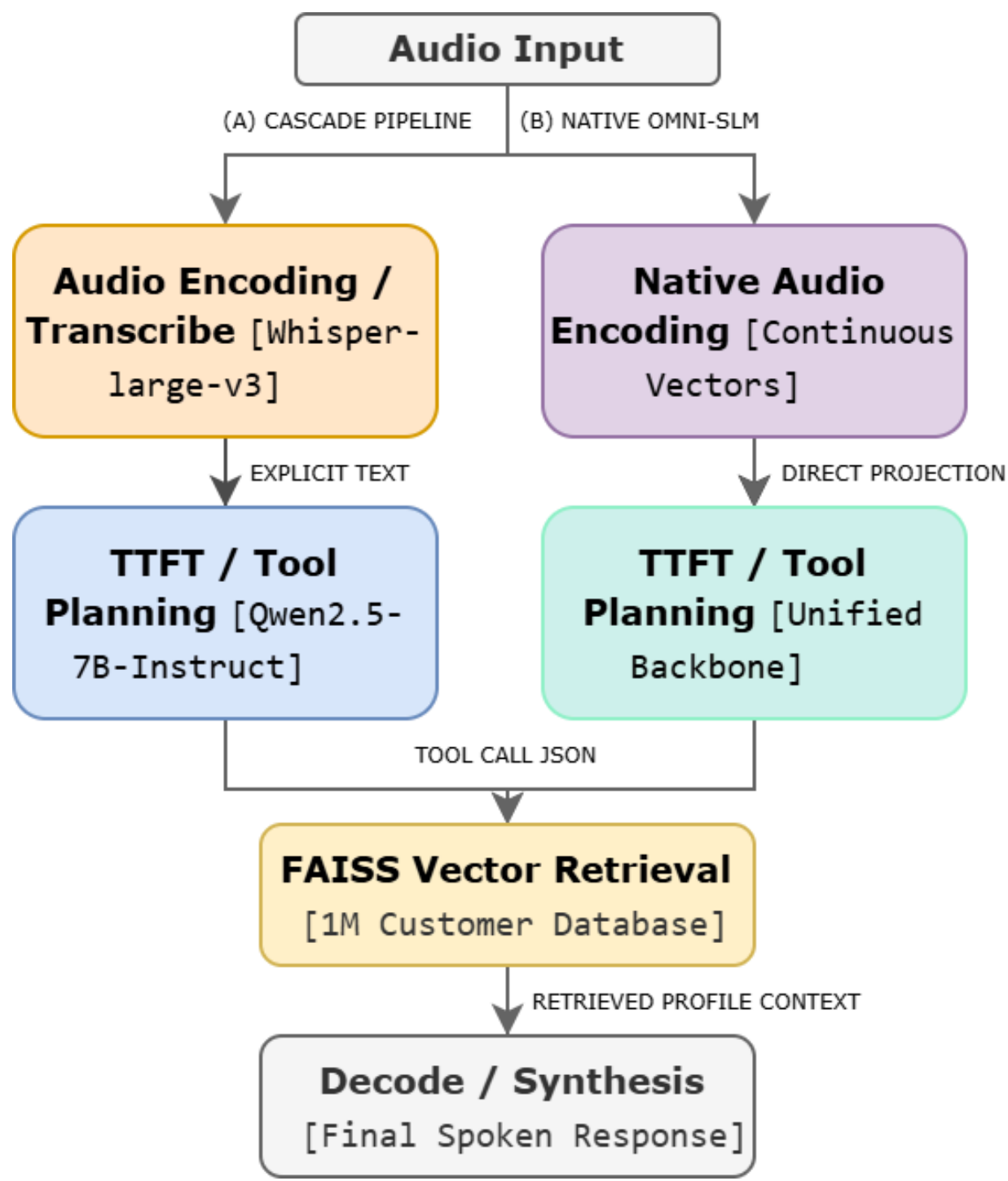
The Problem

- Agents must understand **speech**, **search millions of database profiles**, and **respond dynamically**.

The Goal

- Exact data grounding (zero tolerance for incorrect phone numbers or billing metrics).

The Problem: Latency & Error Cascades



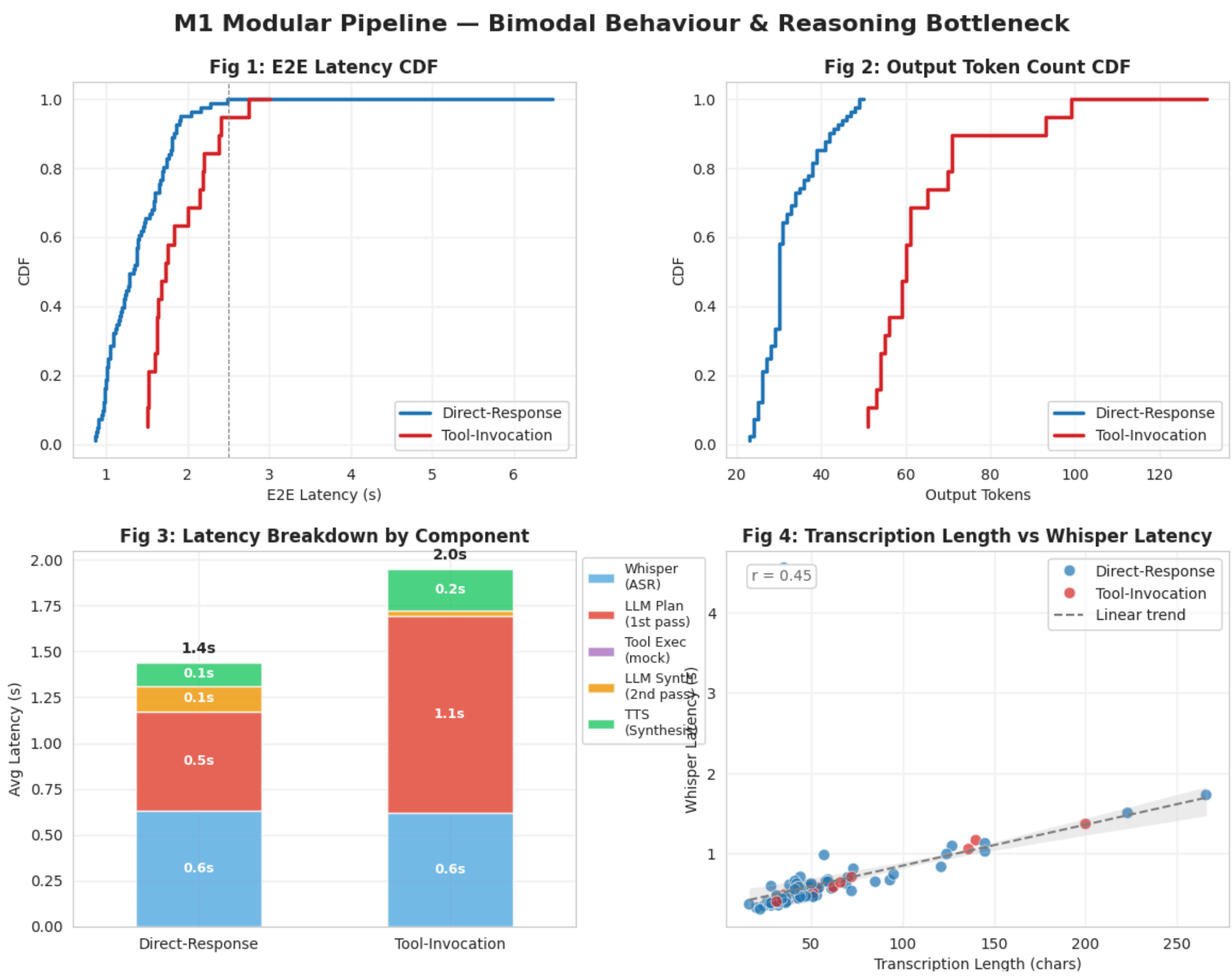
- Compounding Latency:** Traditional systems run ASR, NLU, and LLMs sequentially
- Error Propagation:** If the ASR module mistranscribes a single digit, the downstream LLM cannot recover, leading to lookup failure.
- Hardware Strain:** Deep self-attention over continuous audio frames causes massive memory bandwidth bottlenecks.

Methodology



Results

Template



Template

Metric	Audio Worker	Translation Worker
Number of Requests	50	100
Mean Latency	738.77 ms	49.77 ms
Median Latency	587.22 ms	49.77 ms
P99 Latency	5392.10 ms	66.34 ms
Min Latency	207.22 ms	33.20 ms
Max Latency	9494.38 ms	66.34 ms
Std Dev	1281.90 ms	16.65 ms

Discussion

Template

Conclusion

- Template